WATER RESEARCH 41 (2007) 2004–2014

Available at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/watres

Comparison of control strategies for nitrogen removal in an activated sludge process in terms of operating costs: A simulation study

A. Stare, D. Vrečko, N. Hvala, S. Strmčnik*

Department of Systems and Control, Jožef Stefan Institute, Jamova 39, 1000 Ljubljana, Slovenia

ARTICLE INFO

Article history:
Received 8 June 2006
Received in revised form
19 January 2007
Accepted 23 January 2007
Available online 7 March 2007

Keywords:
Activated sludge process
Wastewater treatment benchmark
Nitrogen removal
Control

ABSTRACT

In this paper several control strategies for nitrogen removal are proposed and evaluated in a benchmark simulation model of an activated sludge process. The goal is to determine which control strategy delivers better performance with respect to plant operating costs. In the study, constant manipulated variables and various PI and feedforward control strategies are tested and compared with predictive control, which uses an ideal process model. The control strategies differ in the information used about the process (number of sensors and sensor location) and in the complexity of the control algorithms. To determine the set-points that yield optimal operating costs, an operational map is constructed for each control strategy. Results of the simulation show that with PI and feedforward controllers almost the same optimal operating costs can be achieved as with more advanced MPC algorithms under various plant operating conditions. More advanced control algorithms are advantageous only in cases where the plant is highly loaded and if stringent effluent fines are imposed by legislation.

© 2007 Elsevier Ltd. All rights reserved.

1. Introduction

Water pollution represents one of the most serious environmental problems due to the discharge of nutrients into receiving waters. Hence, stricter standards for the operation of wastewater treatment plants (WWTPs) have been imposed by authorities. In order to meet these standards, improved control of WWTPs is needed. WWTPs should be controlled in such a way that plant operating costs (OC) are minimized, while effluent standards are still maintained.

Different control algorithms for WWTPs have been introduced over the years. For instance, sufficient nitrification can be maintained by applying a constant aeration flow rate, by control of the dissolved oxygen (DO) level at a pre-selected set-point or by using a variable DO set-point controller based on ammonia concentration in the last aerated reactor of the plant (Ingildsen, 2002; Vrečko et al., 2003). On the other hand, the denitrification process is usually controlled by manipulating the external carbon flow rate or internal recirculation flow rate based on nitrate concentration in the last anoxic reactor or in the last aerobic reactor (Lindberg, 1997; Yuan et al., 2002). In recent years different control algorithms have been proposed, from simple ON/OFF and PI control (Ayesa et al., 2006) to complex model predictive control (MPC) (Steffens and Lant, 1999). Unfortunately, various plant configurations, influent characteristics and evaluation criteria have been used in the assessment of control algorithms. As all of these factors influence the choice of a control strategy, it is difficult to say which control algorithm is the most appropriate with respect to minimal OC and best effluent quality, and whether the implementation of complex control algorithms is really necessary.

*Corresponding author. Tel.: +386 1 47 73 798; fax: +386 1 42 57 009.
E-mail address: aljaz.stare@ijs.si (A. Stare).
0043-1354/$ - see front matter © 2007 Elsevier Ltd. All rights reserved.
doi:10.1016/j.watres.2007.01.029

This paper presents several control strategies for nitrogen removal that were designed and tested on a benchmark simulation model of an activated sludge process (ASP). Control strategies differ in the information used about the process (i.e. the number of sensors and sensor location) and in the complexity of the control algorithms. Constant manipulated variables and PI and feedforward (FF) controllers were tested, as well as an advanced MPC controller, which was used as a reference for the other control strategies. In this way the control strategy that produces optimal performance regarding OC and yields satisfactory removal of nutrients can be found.

The paper is organized as follows: In the following section a benchmark simulation model is presented. Then the applied control strategies are described, followed by set-point analysis. Next, the presented control strategies are assessed and compared in terms of OC at various plant operating conditions. Finally, the most important conclusions are drawn.

2. Materials and methods

2.1. Benchmark simulation model

A benchmark simulation model of an ASP was developed by working group No. 1. within COST Actions 624 and 682. The benchmark is a simulation protocol defining a plant layout, a process model, influent data, test procedures and evaluation criteria (Copp, 2002).

2.1.1. Plant layout and process model

The benchmark represents a pre-denitrification plant with two anoxic and three aerated compartments (Copp, 2002). However, in our case one anoxic and four aerobic reactors were used (Fig. 1) in order to achieve lower effluent ammonia concentrations. The IWA activated sludge model No. 1 (ASM1) is used to describe the biological processes in the reactors. A further description of ASM1 can be found in Henze et al. (2000). The secondary settler is modelled as a non-reactive, 10-layer process with a double exponential settling velocity model proposed by Takács et al. (1991).

2.1.2. Influent data and test procedures

The benchmark influent data include three influent files for three different weather conditions: dry, rain and storm. Each of these files contains 14 days of data at 15-min intervals. To calculate benchmark performance the plant is first run to steady state by simulating the plant with the defined constant influent file over a 150-day period. Then, the plant simulation continues by first applying 14 days of the dry influent weather file, followed by 14 days of dry, rain or storm influent file. The performance of the benchmark is then evaluated for the last 7 days of dynamic data.

2.1.3. Evaluation criteria

Various criteria have been defined in the benchmark (Copp, 2002) to assess the performance of the plant. In this work, OC that combine aeration costs, sludge disposal costs, external carbon dosage costs and effluent fines were used to evaluate control strategies, without taking into account the investment costs for the implementation of the control strategy (sensors, actuators). Within the benchmark, pumping costs are also defined, but were not considered in our case, as all the pumping flow rates were kept constant. The operating costs [€/d] were calculated as follows:

$$OC = \gamma_1 AE + \gamma_2 SP + \gamma_3 EC + EF, \quad (1)$$

where AE is aeration energy, SP is sludge production for disposal, EC is external carbon addition, while EF means effluent fines. The weights γ_1, γ_2 and γ_3 were set in proportion to the weights in the operating cost index (OCI) in the benchmark. The OCI weights (γ_1, γ_2 and γ_3) were defined by the benchmark group and were set according to the relative contribution of AE, SP and EC to operating costs. Based on the experience gained from the WWTPs operation they were chosen to be 1, 5 and 3, respectively. In our case, a rough estimate of average electricity price in EU (0.1 €/kWh) was also taken into account and thus all the weights were multiplied by 0.1. Hence, the final values for γ_1, γ_2 and γ_3 were 0.1 €/kWh, 0.5 €/kg and 0.3 €/kg, respectively.

Average aeration energy (AE [kWh/d]) is calculated for the last 7 days of dynamic data (T). The equation suggested in Copp (2002) was improved by also including the volumes of the aerated reactors (Jeppsson, 2005):

$$AE = \frac{24}{T}\int_{t=7d}^{t=14d}\sum_{i=2}^{5}\left[0.0007 \times K_L a_i(t)^2\left(\frac{V_i}{V_{ref}}\right) + 0.3267 \times K_L a_i(t)\left(\frac{V_i}{V_{ref}}\right)\right]dt, \quad (2)$$

where $K_L a_i$ is the oxygen transfer rate (d^{-1}) in the individual aerated reactor, V_i is the volume of the ith reactor and V_{ref} is 1333 m^3.

Sludge production (SP) for disposal (kg/d) is calculated based on the amount of total suspended solids in the wastage

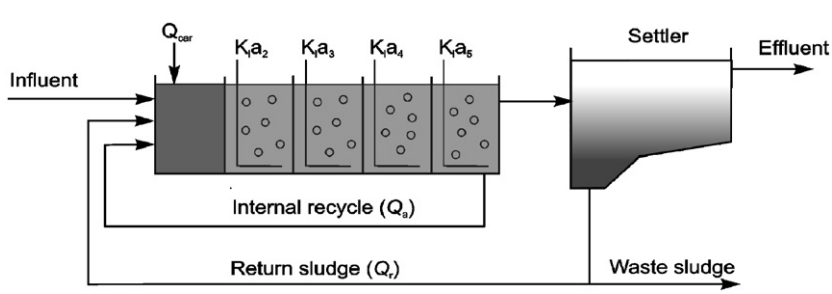

Fig. 1 – Plant layout.

(M_{TSSw}) that is removed from the plant, and the solids accumulated in the system during the last 7 days of operation:

$$SP = \frac{1}{T}(M_{TSSw} + M_{TSS\ system}(14d) - M_{TSS\ system}(7d))$$
$$M_{TSS\ system} = M_{TSS\ reactors} + M_{TSS\ settler}$$
$$M_{TSSw} = 0.75 \int_{t=7d}^{t=14d} [X_{S,w}(t) + X_{BH,w}(t) + X_{BA,w}(t) + X_{P,w}(t) + X_{I,w}(t)]Q_w(t)dt. \quad (3)$$

The amount of solids in the system ($M_{TSSsystem}$) is equal to the amount of solids in the reactors ($M_{TSSreactors}$) and the amount of solids in the settler ($M_{TSSsettler}$). M_{TSSw} is calculated from $X_{S,w}$, $X_{BH,w}$, $X_{BA,w}$, $X_{P,w}$ and $X_{I,w}$, which represent concentrations of slowly biodegradable substrate, active heterotrophic and autotrophic biomass, particulate products arising from biomass decay and particulate inert organic matter in the waste sludge, respectively.

External carbon (EC) addition [kg/d] is calculated as the average external carbon mass flow:

$$EC = \frac{COD_S}{T \times 1000} \int_{t=7d}^{t=14d} Q_{car}(t)dt, \quad (4)$$

where COD_S is the carbon source concentration (4×10^5 mg/l) and Q_{car} is the carbon flow rate (m^3/d).

Effluent fines (EF) are usually paid in proportion to the discharge of pollution into receiving waters. In our study, only effluent ammonia (S_{NHeff}) and total nitrogen (S_{TNeff}) were considered in the calculation of effluent fines. The reason for this is that the controllers applied in this paper do not have much impact on effluent COD and TSS, which are usually included in the effluent fines. It should be noted that omitting COD and TSS from the EF calculations results in a greater relative difference between the controllers, while the absolute difference between the controllers remains approximately the same. However, as it is quite difficult to remove all the nitrogen components in the benchmark plant, the effluent nitrogen fines are much higher compared to the fines for discharging COD and TSS, which means that also a relative difference between the controllers is not much different despite omitting COD and TSS from the EF calculations.

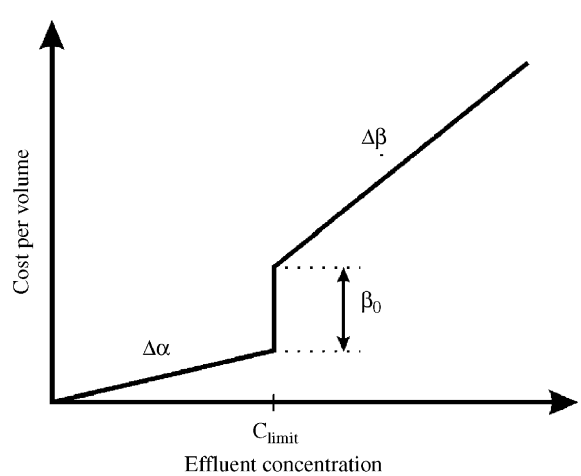

Fig. 2 – Cost function for effluent fines.

The cost function for EF which was used in our case is shown in Fig. 2 (Carstensen, 1994; Vanrolleghem et al., 1996). $\Delta\alpha$ and $\Delta\beta$ are costs per kilogram of discharge below and above the effluent limit concentration C_{limit}, while β_0 is the cost for exceeding the effluent limit. A complete mathematical description of the cost functions is:

$$Cost(t) = \begin{cases} \Delta\alpha C_{eff} Q_{eff} & \text{if } C_{eff} \leqslant C_{limit}, \\ \Delta\alpha C_{limit} Q_{eff} + \beta_0 Q_{eff} + \Delta\beta(C_{eff} - C_{limit})Q_{eff} & \text{if } C_{eff} > C_{limit}, \end{cases} \quad (5)$$

where C_{eff} and Q_{eff} are the effluent concentration and flow rate, respectively. In our case the following values of cost function parameters are used. When effluent ammonia and total nitrogen are below the discharge limit, the costs of discharge are $\Delta\alpha_{NH} = 4$ €/kg and $\Delta\alpha_{TN} = 2.7$ €/kg, respectively. When the discharge limits are exceeded, the costs of discharge increase threefold ($\Delta\beta_{NH} = 12$ €/kg and $\Delta\beta_{TN} = 8.1$ €/kg). The costs of exceeding the discharge limits are $\beta_{0,NH} = 2.7$ €/1000 m^3 and $\beta_{0,TN} = 1.4$ €/1000 m^3. Discharge limits for ammonia ($S_{NHlimit}$) and total nitrogen ($S_{TNlimit}$) were set to 4 and 12 mg/l, respectively, in our case. Note that ammonia is penalized twice, as ammonia and also as part of total nitrogen.

2.2. *Applied control strategies*

In this paper the following manipulated variables were used in the control strategies: external carbon flow rate (Q_{car}), DO set-point (So_{set}) and oxygen transfer rate (K_La), which is related to the airflow rate. Other manipulated variables such as internal recycle flow rate (Q_a), external recycle flow rate (Q_r) and waste sludge flow rate (Q_w) were set to the constant values of 55,338, 18,446 and 300 m^3/d, respectively.

Boundaries of the manipulated variables were the same as in the benchmark. The maximum external carbon flow rate was limited to 5 m^3/d, with a COD concentration of 4×10^5 mg/l, while the oxygen transfer rate (K_La) values were limited between 0 and 360 d^{-1}. The upper bound of the oxygen set-point was limited to 3 mg/l in order to reduce unnecessary aeration of the aerobic reactors. Because of the constraints of the manipulated variables, anti-windup protection, similar to that used in the benchmark, was applied in the PI controllers. A short description of the applied control strategies is given in the following paragraphs.

2.2.1. *Constant manipulated variables*

The most basic control strategy, which is still used in practice, does not contain feedback loops. It simply uses constant manipulated variables. In our case, constant K_La and Q_{car} were applied as shown in Fig. 3a.

2.2.2. *Oxygen PI control*

Oxygen PI control (Fig. 3b) uses four PI controllers that control DO concentrations in the aerobic reactors at the desired set-point by manipulating the K_La values. This control strategy needs four additional DO sensors, which are the most commonly used in practice for online control or monitoring. Since the DO sensors are proven to be robust and reliable with relatively fast response times, the benchmark DO sensors are

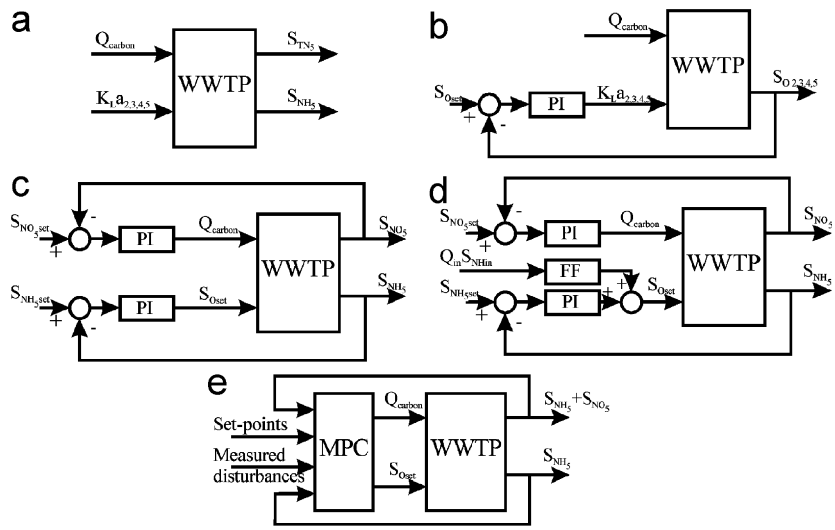

Fig. 3 – Control structures: (a) constant manipulated variables, (b) oxygen PI control, (c) nitrate and ammonia PI control, (d) nitrate PI and ammonia FF-PI control, and (e) model predictive control

assumed to be ideal with no delay and noise. In full scale implementation the DO controller is cascaded with the airflow controller in order to avoid nonlinearities in valve operation. The inner controller that receives the airflow rate set-point and controls valve position is not used in the benchmark since it does not influence the final results due to fast valve response.

2.2.3. *Nitrate and ammonia PI control*

In this control strategy, two additional control loops are applied (Fig. 3c). In the first loop, nitrate concentration in the last aerobic reactor (S_{NO_5}) is controlled with the PI controller that manipulates the external carbon flow rate (Q_{car}). In the second loop, ammonia concentration in the last aerobic reactor (S_{NH_5}) is controlled with a cascade controller. The outer PI controller in the cascade control adjusts the DO set-point value (So_{set}) based on the desired and actual S_{NH} values, and the inner PI controllers manipulate K_La values based on the desired and actual DO values. To apply this control strategy in practice, additional sensors for ammonia and nitrate concentrations are needed. These sensors are not as reliable and responsive as DO sensors, for example. Hence, more realistic S_{NH} and S_{NO} sensor models are included within the benchmark. Both sensors are assumed to have a time delay of 10 min with normally distributed white zero-mean noise. The standard deviation of the S_{NO} and S_{NH} sensors is set to 0.1 and 0.2 mg/l, respectively.

2.2.4. *Nitrate PI and ammonia feedforward-PI control*

The idea of FF control is to act on the process when disturbances appear and before they cause deterioration of the effluent quality. In the proposed strategy, the ammonia cascade PI controller was upgraded with a FF control that uses influent flow rate (Q_{in}) and influent ammonia concentration (S_{NHin}) as measurable disturbances. The control scheme of nitrate PI and ammonia FF-PI control is shown in Fig. 3d.

The DO set-point is in this case calculated as the sum of PI and FF control:

$$So_{set} = So_{PI} + So_{FF}, \tag{6}$$

where So_{PI} is the output of the ammonia PI controller and So_{FF} is the output of the FF controller. The response of the FF controller is proportional to the product of S_{NHin} and Q_{in} and was chosen to be non-zero only during high influent loads:

$$So_{FF} = \begin{cases} 0 & \text{if } Q_{in}S_{NHin} \leqslant n_{load}, \\ k(Q_{in}S_{NHin} - n_{load}) & \text{if } Q_{in}S_{NHin} > n_{load}. \end{cases} \tag{7}$$

High influent flow rates reduce hydraulic retention times and therefore it is important to increase DO concentrations fast enough to counteract the influent disturbances. During low influent loads the PI controller is sufficiently fast to reject small influent disturbances, and hence the FF part can be switched off. The FF part is switched on/off when the influent nitrogen load ($Q_{in}S_{NHin}$) is higher/lower than the threshold value n_{load}, which was in our case set to a relatively high value of 1000 kg/d (e.g., the average influent nitrogen load in the benchmark is approximately 550 kg/d). The proportional factor k in (7) was set to a value of 2×10^{-2} (mgO_2 d)/(kgN l). In this way the So_{set} is increased by 2 mg/l when the influent load reaches 1100 kg/d. For this control strategy the same S_{NH} and S_{NO} sensor models were applied as in the case of nitrate and ammonia PI control.

2.2.5. *Model predictive control*

In order to determine if a more advanced control strategy can further reduce OC and improve effluent quality, a multivariable and nonlinear MPC was implemented. The idea of MPC is to calculate a control sequence by minimizing a certain cost function. The control sequence is calculated at each time step based on set-points, a process model, measured disturbances and outputs (Maciejowski, 2002). In our case, the whole benchmark model with perfect measurements (also for ammonia and nitrate) was used as the process model of the predictive controller, and all influent

disturbances were assumed to be known in advance over the future prediction horizon. This "ideal" control algorithm was used as a reference for the other control algorithms presented above and should represent the upper limit of what can be achieved by predictive control.

The general cost function used in the MPC algorithm can be written as follows:

$$J = \sum_{i=1}^{H_p}(\hat{y}(k+i) - r(k+i))^T Q(\hat{y}(k+i) - r(k+i)) + \sum_{i=1}^{H_u}\Delta u(k+i)^T R_{\Delta u}\Delta u(k+i) + \sum_{i=1}^{H_u}(u(k+i) - u_0)^T R_u(u(k+i) - u_0) + \rho\sum_{i=1}^{H_p}\varepsilon(k+i), \quad (8)$$

where k denotes the sampling instants, $\hat{y}(k+i)$ is a vector of the predicted output values, $r(k+i)$ is a vector of the future set-point values, $\Delta u(k+i)$ is a vector of the future input changes, $u(k+i)$ is an input vector, $\varepsilon(k+i)$ is a slack variable, H_p is the prediction horizon, H_u is the control horizon, Q is a weighting matrix to penalize the error between the predicted process output and the set-point, $R_{\Delta u}$ is a weighting matrix to penalize changes in the control signal, R_u is a weighting matrix to penalize deviations of the input vector from the desired steady-state value (u_0), while ρ is a weight to penalize soft constraint violations.

In our case, ammonia (S_{NH_5}) and the sum of ammonia and nitrate concentrations ($S_{NH_5} + S_{NO_5}$) in the last aerobic reactor were used as controlled variables, while the manipulated variables were DO set-point (So_{ref}) and external carbon flow rate (Q_{car}). Input and output vectors can be written as follows:

$$y = \begin{bmatrix} S_{NH_5} \\ S_{NO_5} + S_{NH_5} \end{bmatrix}, \quad u = \begin{bmatrix} So_{ref} \\ Q_{car} \end{bmatrix}. \quad (9)$$

The control scheme of the MPC is shown in Fig. 3e.

2.3. *Tuning of controllers*

2.3.1. *Tuning of PI controllers*

Parameters of PI controllers were tuned from the step response experiments using the internal model control (IMC) tuning rules (Olsson and Newell, 1999). The values of the parameters of PI controllers are shown in Table 1.

2.3.2. *Tuning of the MPC*

Parameters of the prediction and control horizon, the weights and the set-point trajectory of the MPC affect the closed-loop behaviour of the plant. The weights in the cost function may be set according to the economic objectives of the plant, but usually they are adjusted in such a way that satisfactory control performance is achieved (Maciejowski, 2002). In our case, the parameters were tuned based on experience gained from the simulations and from the tuning rules presented in Maciejowski (2002). Since the dynamics of nutrient removal processes are on a time scale of hours, the prediction horizon was set to 1.5 h, which is 6 time steps (each time step was 15 min). In order to simplify the calculation of the input sequences, only one control move Δu was optimized at each sampling instant ($H_u = 1$), i.e. the control signal was constant during the prediction horizon. The weighting matrix Q was set to diag[10 0] so that only the deviation of S_{NH_5} from the desired set-point was penalized. A soft constraint on S_{NH_5} was also used to additionally penalize concentrations above the effluent limit. This soft constraint was implemented by using a so-called "slack variable", which is defined in such a way that it is non-zero only when the constraint is violated, i.e. when S_{NH_5} values exceed 4 mg/l. The other controlled variable (S_{NH_5}+S_{NO_5}) was considered in the cost function only with the soft constraint, which was set to 10 mg/l. In this way the aim was to keep S_{NH_5}+S_{NO_5} below 10 mg/l, which could actually be achieved by keeping the value of S_{NH_5}+S_{NO_5} at the desired set-point of 10 mg/l. However, when S_{NH_5}+S_{NO_5} is below 10 mg/l, no effort is put into increasing S_{NH_5}+S_{NO_5} concentration, as would be the case with set-point tracking. To prevent external carbon from being unnecessarily dosed during low loads (i.e. when the S_{NH_5}+S_{NO_5} concentration is below 10 mg/l) the weighting matrix R_u was set to diag[0 1]. In addition, changes of the manipulated variables were penalized with weighting matrix $R_{\Delta u}$, which was set to diag[1 1]. This has the effect of reducing the oscillations of the control signals. The parameters of the MPC can be summarized as follows:

$$Q = \begin{bmatrix} 10 & 0 \\ 0 & 0 \end{bmatrix}, R_{\Delta u} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, R_u = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad u_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \rho = 10, H_p = 6, H_u = 1. \quad (10)$$

Table 1 – Parameters of PI controllers

	Oxygen PI controller	Ammonia PI controller	Nitrate PI controller
K_p	100	−0.3	−0.7
T_i	0.01 d	0.2 d	0.1 d

3. Results

3.1. *Set-point analysis*

The performance of the plant strongly depends on the selected controller set-points. In order to study the impact of the controller set-points on plant performance, a number of dynamic simulations with different set-point values were performed. With such an analysis, operational maps were constructed (see Figs. 4–6) to determine optimal operating points with respect to OC for each control strategy. In the figures the maximum values of the effluent ammonia and effluent total nitrogen concentrations are also plotted in order to see how successfully the controllers reject influent disturbances. The obtained optimal set-points with respect to OC are shown in Table 2.

3.2. *Comparison of control strategies at optimal set-point values*

Each control strategy was simulated using the optimal set-point values given in Table 2. The results of simulations from

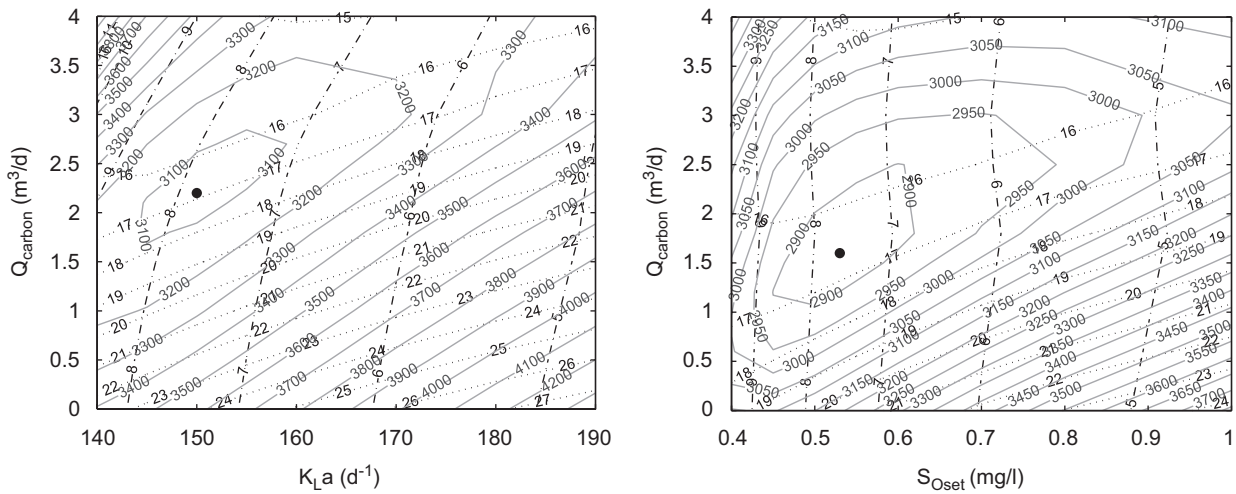

Fig. 4 – Operational map for constant manipulated variables (left) and for oxygen PI control (right). Operating costs (solid line), maximum effluent total nitrogen concentration (dotted line), maximum effluent ammonia concentration (dash–dotted line). The dot indicates the minimum of operating costs.

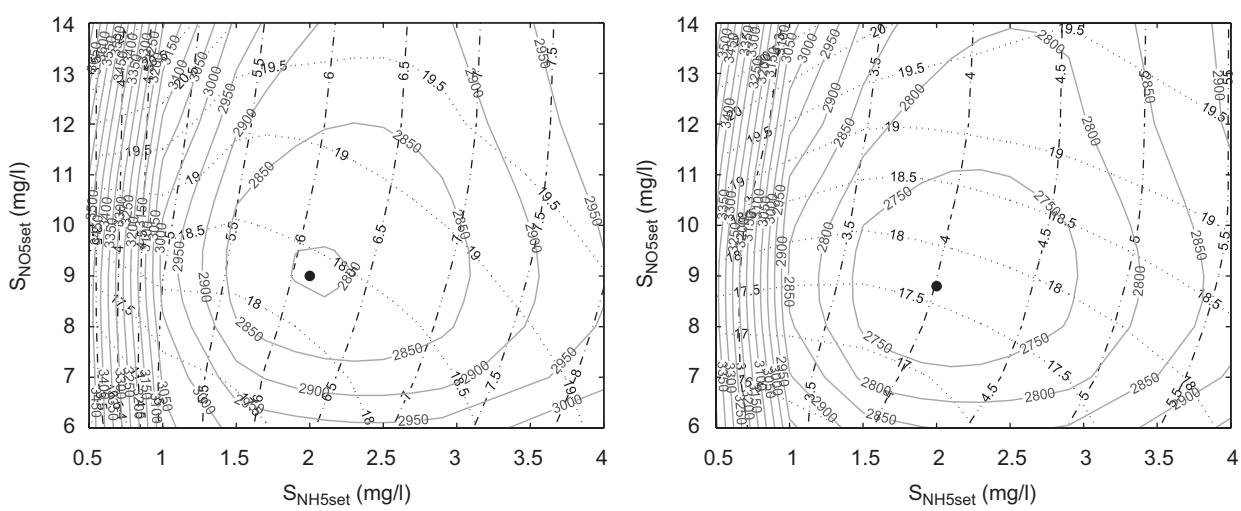

Fig. 5 – Operational map for nitrate and ammonia PI control (left) and for nitrate and ammonia feedforward-PI control (right). Operating costs (solid line), maximum effluent total nitrogen concentration (dotted line), maximum effluent ammonia concentration (dash–dotted line). The dot indicates the minimum of operating costs.

the 8th to the 10th day of influent data are shown in Figs. 7–9. From Fig. 7 (left) it can be seen that when constant manipulated variables are used, the removal of ammonia is not efficient. In this case the aerobic reactors are under-aerated during high influent loads (Fig. 8, left), leading to poor ammonia removal, and over-aerated during low loads, leading to unnecessarily high DO concentrations and energy consumption in these periods (Fig. 8, right). With an oxygen PI controller that controls DO concentrations at the desired constant set-point, the plant is better controlled with respect to energy consumption, but still under-aerated during high influent loads. Considerably better removal of ammonia is achieved by using control strategies where ammonia concentrations are controlled (Fig. 7, left). In these cases, the ammonia controller increases DO concentrations during high load periods (see Fig. 8, right) to enhance nitrification and to remove ammonia peaks more successfully, while during low loads it decreases the DO concentrations to save aeration energy. In addition, low DO set-points also enable simultaneous denitrification in the aerobic reactors, leading to lower total nitrogen concentrations (Fig. 7, right). In the case of ammonia PI and ammonia FF-PI control, the maximum effluent S_{NH} peak is reduced to 6 and 4 mg/l, respectively. The differences between these two cases with respect to ammonia peak removal are quite significant. With an ammonia PI controller the DO concentrations are not increased during high influent loads until S_{NH_5} is high. Hence, while ammonia concentrations in the first three aerobic reactors are high, considering the propagation of the influent load through the reactors, the nitrification rate is still low

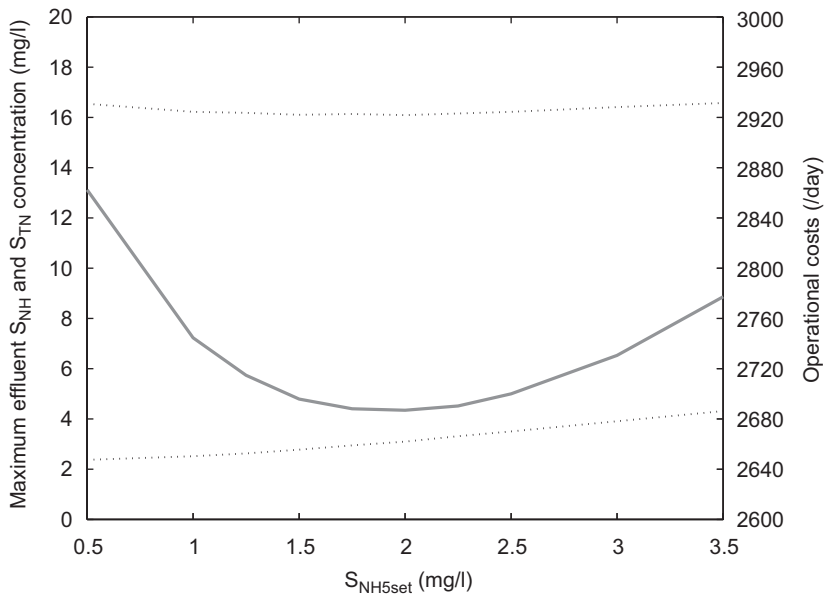

Fig. 6 – Operational map for MPC. Operating costs (solid line), maximum effluent total nitrogen concentration (dotted line), maximum effluent ammonia concentration (dash–dotted line).

Table 2 – Optimal set-point values for each control strategy

	Optimal values of set-points
Constant manipulated variables	$K_La = 150\,d^{-1}$, $Q_{car} = 2.2\,m^3/d$
Oxygen PI control	$So_{set} = 0.53\,mg/l$, $Q_{car} = 1.6\,m^3/d$
Nitrate and ammonia PI control	$S_{NH_5set} = 2.0\,mg/l$, $S_{NO_5set} = 9.0\,mg/l$
Nitrate PI and ammonia FF-PI control	$S_{NH_5set} = 2.0\,mg/l$, $S_{NO_5set} = 8.8\,mg/l$
MPC	$S_{NH_5set} = 2.0\,mg/l$

because of low DO concentrations. With ammonia FF-PI control, a better reduction of ammonia peaks is achieved because the DO concentrations are increased more than 1 h earlier (Fig. 8, right), which leads to better removal of S_{NH}. In the case of MPC the algorithm increases DO concentrations even a little earlier and hence further improvements with respect to ammonia removal are obtained.

Even though the upper limit for DO concentration was 3 mg/l, none of the controllers could manage to remove the effluent ammonia peaks completely. The problem is that the aeration volumes of the benchmark are too small for the N load applied to the plant, and should be increased in order to achieve complete ammonia removal. An improvement in ammonia removal could be also achieved by increasing the sludge age (i.e. reducing the waste sludge flow rate). However, it should be mentioned that the default benchmark value of Q_w was already reduced from 385 to 300 m^3/d in our case. Therefore, even lower Q_w value can cause a significant increase in the effluent suspended solids concentrations as a result of higher mixed liquor suspended solids (MLSS) concentration and thus increased solids loading to the settler. This can be especially evident during heavy rainfall and storm events as these have an important effect on the hydraulics of a treatment plant and may disturb the final settling to such an extent that sludge wash-out occurs.

Comparison of carbon addition flow rates (Fig. 9) shows that with the constant carbon addition flow rates (in the case of constant manipulated variables and oxygen PI control) similar problems occur as with constant aeration, i.e. too much external carbon is unnecessarily added during low influent loads, while too little carbon is added during high influent loads. External carbon can be dosed more efficiently with a PI controller that adds external carbon only when the nitrate concentration in the fifth reactor is high in order to enhance denitrification. The MPC controller starts to change the external carbon dosing much earlier than the nitrate PI controller, and therefore a better reduction of effluent total nitrogen peaks is achieved (Fig. 7, right). Fig. 7 shows that during high influent loads the removal of S_{TNeff} peaks is much better with constant manipulated variables and oxygen PI control than with nitrate PI control. Namely, in the first two cases, low DO concentrations during high influent loads enable simultaneous denitrification in aerobic reactors, leading to lower S_{TNeff}. However, during low influent loads the S_{TNeff} concentrations are much higher in these cases because of high DO concentrations.

Control strategies were also evaluated with the criteria described in Section 2.1.3. The obtained values of the evaluation criteria are given in Table 3. The control strategy with constant manipulated variables gives the largest OC and is used as a reference case for comparison. It can be seen that with each control strategy introduced the OC are further reduced. An important factor contributing to lower OC are lower effluent fines, which constantly decrease with each control strategy introduced. A reduction in OC is also

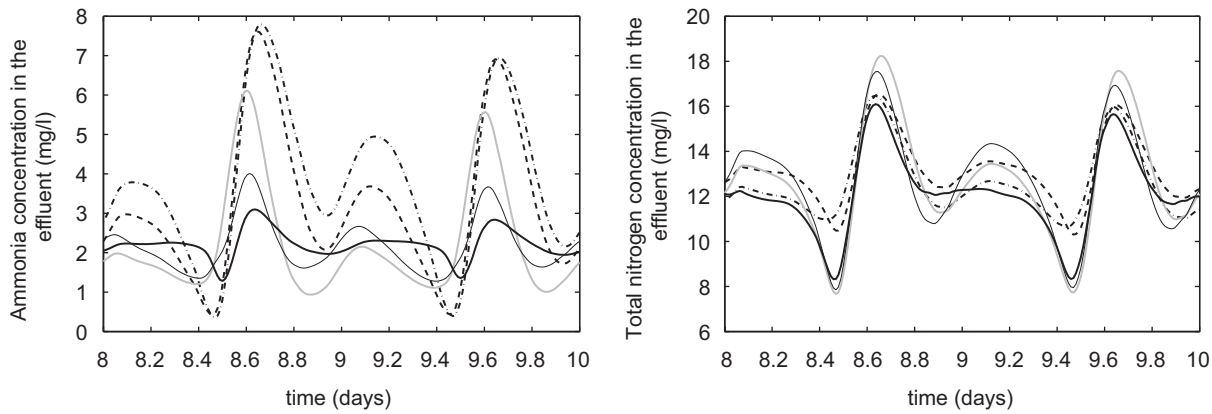

Fig. 7 – Comparison of effluent ammonia (left) and total nitrogen (right) concentrations. Constant manipulated variable (dash–dotted line), oxygen PI control (dashed line), nitrate and ammonia PI control (solid grey line), nitrate PI and ammonia FF-PI control (thin solid black line), MPC (solid black line).

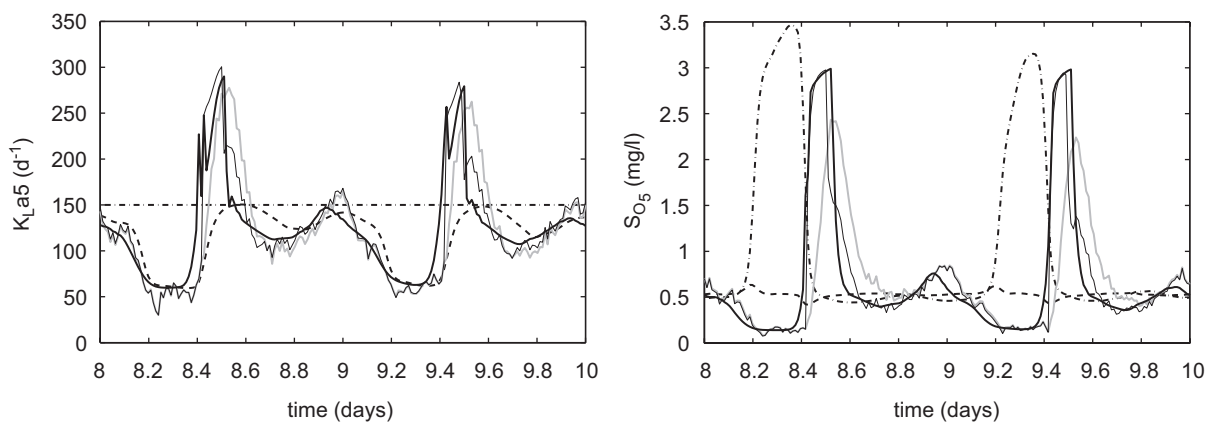

Fig. 8 – Comparison of K_La (left) values and oxygen concentrations (right) in the fifth reactor. Constant manipulated variable (dash–dotted line), oxygen PI control (dashed line), nitrate and ammonia PI control (solid grey line), nitrate PI and ammonia FF-PI control (thin solid black line), MPC (solid black line).

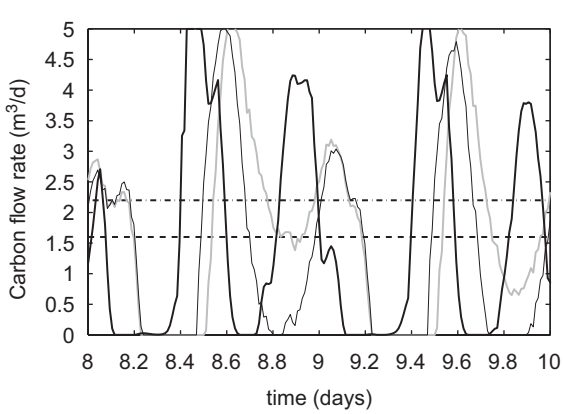

Fig. 9 – Comparison of external carbon flow rates. Constant manipulated variable (dash–dotted line), oxygen PI control (dashed line), nitrate and ammonia PI control (solid grey line), nitrate PI and ammonia FF-PI control (thin solid black line), MPC (solid black line).

achieved because of lower aeration costs, which decrease with the introduction of oxygen control, but slightly increase with ammonia controllers because of the greater energy needed for better ammonia removal. Similarly, the costs of carbon addition greatly decrease with the introduction of nitrate controllers, but slightly increase in the case of MPC because more carbon is needed for better total nitrogen removal.

OC is reduced by about 7% when oxygen PI control is used, amounting to around 203 € per day (or 74,000 € per year), which is reasonably high taking into account that only simple oxygen PI controllers have to be used. Further cost reductions of 2.5% (74 € per day) are achieved by using nitrate and ammonia PI control. With nitrate PI control and ammonia FF-PI control, OC are reduced by an additional 3% (85 € per day). In the case of MPC, only a slight improvement in OC is achieved (around 1% or 25 € per day). In this case, effluent

Table 3 – Evaluation criteria for different control strategies

	Constant manipulated variables	Oxygen PI control	Nitrate and ammonia PI control	Nitrate PI and ammonia FF-PI control	MPC
Aeration costs (€/d)	576	512 (−11.1%)	537 (−6.8%)	541 (−6.1%)	538 (−6.6%)
Sludge disposal costs (€/d)	1258	1227 (−2.5%)	1215 (−3.4%)	1201 (−4.5%)	1220 (−3.0%)
Carbon costs (€/d)	264	192 (−27.3%)	148 (−44.0%)	123 (−53.4%)	152 (−42.4%)
Effluent fines (€/d)	976	940 (−3.7%)	897 (−8.1%)	847 (−13.2%)	777 (−20.4%)
Operating costs (€/d)	3074	2871 (−6.6%)	2797 (−9.0%)	2712 (−11.8)	2687 (−12.6%)

Table 4 – Operating costs for control strategies at various plant conditions

Operating costs (€/d) with	Constant manipulated variables	Oxygen PI control	Nitrate and ammonia PI control	Nitrate PI and ammonia FF-PI control	MPC
Increased S_{Sin} (+15%)	3050	2827 (−7.3%)	2744 (−10.0%)	2659 (−12.8%)	2643 (−13.3%)
Decreased S_{Sin} (−15%)	3148	2958 (−6.0%)	2857 (−9.2%)	2772 (−11.9%)	2739 (−13.0%)
Increased S_{NHin} (+15%)	3693	3509 (−5.0%)	3328 (−9.9%)	3189 (−13.6%)	3142 (−14.9%)
Decreased S_{NHin} (−15%)	2754	2535 (−8.0%)	2357 (−14.4%)	2324 (−15.6%)	2325 (−15.6%)
Increased Q_{in} (+15%)	4310	3893 (−9.7%)	3757 (−12.8%)	3589 (−16.7%)	3527 (−18.2%)
Decreased Q_{in} (−15%)	2505	2191 (−12.5%)	2023 (−19.2%)	1998 (−20.2%)	1986 (−20.7%)
Storm event	3360	3150 (−6.3%)	3061 (−8.9%)	2975 (−11.5%)	2955 (−12.1%)
Rain event	3293	3211 (−2.5%)	3045 (−7.5%)	2956 (−10.2%)	2911 (−11.6%)
Decreased T (12 °C)	3549	3694 (+4.1%)	3365 (−5.2%)	3251 (−8.4%)	3143 (−11.4%)
Increased T (18 °C)	2830	2607 (−7.9%)	2416 (−14.6%)	2382 (−15.8%)	2357 (−16.7%)
Average OC (€/d)	3249	3058 (−5.9%)	2895 (−10.9%)	2810 (−13.5%)	2773 (−14.7%)

fines are reduced considerably, but OC increase because of the greater addition of external carbon.

3.3. *Performance of controllers at different influent and temperature conditions*

Besides the comparison of the best possible performance of the controllers, it is of great interest to study how the controllers perform under different operating conditions. In reality, WWTPs are subjected to different seasonal effects (i.e. temperature changes, influent composition and concentrations variations, different sludge properties, etc.), which means that controller set-points that were optimal at certain plant conditions are suboptimal under other conditions. Therefore, this section investigates the performance of different controllers if the plant loading conditions and temperature change. To change the plant loading conditions, S_{NHin}, S_{Sin} and Q_{in} were multiplied by a factor 1.15 or 0.85 during the last 7 days of dry influent data, thus increasing or decreasing the plant's average influent load and its dynamics. In addition, the rain and storm conditions as defined in the benchmark were studied as well. Temperature was changed to 12° and 18° for the whole simulation period by adjusting the activated sludge kinetic parameters (μ_H, b_H, μ_A, b_A, k_h and k_a) through an Arrhenius function. At 15° the kinetic parameters are equal to the default benchmark parameters. In the analysis the controller set-points were set to the same values obtained under the default benchmark conditions (Table 2).

In Table 4 the control algorithms are compared with respect to OC for different plant conditions. For each control strategy, the percentage change of OC compared to the case of constant manipulated variables is also calculated. The percentages are shown for each individual change in plant operating conditions, as well as for the average OC calculated from all cases studied (bottom line in Table 4). In virtually all cases the results show that OC decreased with the introduction of new control strategies, the only exception being oxygen PI control at 12 °C, where OC increased compared to constant manipulated variables because of a low DO set-point. The differences between nitrate and ammonia PI control, nitrate PI and ammonia FF-PI, and MPC are more obvious at higher influent loads than at lower loads, while the improvements with MPC are rather small. From the study it can be concluded that more advanced controllers result in lower OC over a wider range of plant operating conditions and also when the controller set-points are not chosen optimally for a given plant's operating conditions. From Table 4 it can be seen that the reduction of OC depends on plant operating conditions (influent load, temperature), but on average the reduction of costs with more advanced control strategies is even slightly higher than at the default benchmark conditions in Table 3. This can be explained by the fact that with the ammonia and nitrate controllers the change of operating

conditions acts as a disturbance and the controllers themselves adjust the low-level set-points (DO set-point, K_La, external carbon flow rate), while in the case of constant manipulated variables and oxygen control these variables are kept constant despite process changes. In these two cases the choice of optimal set-points is more sensitive to process changes.

4. Discussion

The results presented here show that with nitrate PI and ammonia FF-PI control almost the same OC can be achieved as with advanced MPC algorithms. Only when the plant is highly loaded would an MPC algorithm be advantageous. It should be emphasized that the MPC used in the study was ideal with respect to the exact process model, all measurements available and perfect knowledge of the disturbances. It can be expected that any MPC implementation in a real system would result in smaller differences with respect to OC and effluent quality. Hence, it appears that additional information about the plant, along with an advanced multivariable and nonlinear control algorithm, would not improve the operation of the plant to such an extent that the investment in buying additional sensors and implementing more complex control algorithms would pay off. The improvement is thus more related to control structure, i.e. where the sensors and the actuators are located, than to the control algorithms. Similar conclusions were also arrived at in a simulation study by Lukasse and Keesman (1999), where a simple feedback controller based on easily implemented rules was designed to closely imitate the optimal operating strategy for an alternating ASP. However, it should also be kept in mind that the results presented are based on a simulation model, which is only an approximation of the real process. Therefore, full-scale experiments would also be needed in order to additionally support the conclusions.

In general, the optimal OC for each control strategy strongly depends on established effluent fines, i.e. selected parameters of $\Delta\alpha$, $\Delta\beta$ and β_0. In our case the cost of discharge increased threefold each time the discharge limits were exceeded (Fig. 2), which favours control algorithms that keep the effluent concentrations away from their defined limits. However, if a constant effluent fine per kg of pollutant discharged over the complete effluent concentration range is used ($\beta_0 = 0$, $\Delta\beta = \Delta\alpha$), it turns out that smaller improvements can be obtained with advanced control algorithms. For example, under the default benchmark conditions the optimal OC and percentage improvement compared to constant manipulated variables are 2691 €/d, 2613 €/d (−2.9%), 2559 €/d (−4.9%), 2536 €/d (−5.8%) and 2525 €/d (−6.2%) for the constant manipulated variables, oxygen PI control, nitrate and ammonia PI control, nitrate PI and ammonia FF-PI control, and MPC, respectively. The results also indicate that, in this case, the optimal set-points lie in a region where Q_{car} is zero, and thus due to softer effluent fines external carbon should not be added in order to minimize OC (carbon costs, sludge production costs as well as aeration costs). On the other hand, heavier effluent fines would favour advanced control algorithms even more. While a study such as this one can show what is possible to achieve in the operation of WWTPs, the actual implementation of advanced control strategies strongly depends on the effluent fines imposed by legislation. For sensitive discharge points it is important to prevent high peak values from occurring, and therefore it is better to implement advanced control algorithms. However, for less sensitive water bodies, peak discharge will not be so important and hence it is probably sufficient to use simple control algorithms.

In the study the same sensors that are defined within the benchmark were applied. However, a number of sensors with different performance characteristics are commercially available, which means that in reality it is vital to use sensors with short response times and low measurement noise as the control performance strongly depends on these sensor characteristics. For example, if S_{NH} and S_{NO} sensors with even longer response times and higher measurement noise were used, the difference between the oxygen PI control algorithm and the nitrate and ammonia PI control algorithm would be even smaller, while the usage of FF controller would be more favourable in this case. On the other hand, S_{NH} and S_{NO} PI control algorithm would achieve almost optimal performance regarding OC if sensors with shorter response times and lower measurement noise were used as the proportional gain of PI controllers can be increased, which leads to faster controller response and hence better rejection of influent disturbances.

It should also be noted that the investment and maintenance costs are higher in the case of complex control algorithms because of a large number of actuators and sensors required. In reality actuators and sensors are also subjected to failure, which means that control algorithms that use a large number of actuators and sensors are also more vulnerable to malfunctions. These kinds of problems should be solved with additional fault detection algorithms that detect errors and failures and that initiate proper steps to correct them. As detection and accommodation of failures takes time the advanced control algorithms will consequently provide optimal behaviour for a shorter period of time. In our study, the evaluation of control strategies based on such malfunctions was not considered.

From the study it can be seen that simple feedback controllers are often sufficient to reach improved plant performance with respect to OC, while the MPC algorithm could not deliver much better results despite the fact that additional information about the process was used. Hence, it is unlikely that with the existing manipulated variables (K_La, So_{set}, Q_{car}) one can achieve much greater reduction of OC. Some further improvements could be expected with the use of additional manipulated variables such as internal recycle flow rate, external recycle flow rate and waste sludge flow rate. However, the main reason that greater reduction of OC cannot be achieved is due to the plant configuration with relatively small total reactor volumes for a given influent load. To avoid such a problem, the integrated process design and control may be used, which aims at identifying design decisions that would potentially generate and inherit possible trouble to the dynamic performance (Seferlis and Georgiadis, 2004). In our case only one anoxic reactor within the given total reactors volume was used, which limits the denitrification process. On the other hand, if two anoxic reactors had been used, the nitrification process would have been limited. Therefore, the introduction of the aeration

volume control (i.e., the control of the distribution between the aerobic and anoxic reactor volumes) may be an interesting subject for further research.

5. Conclusions

In this paper several control strategies for nitrogen removal in an ASP were evaluated using a benchmark simulation model. The aim of the study was to investigate the differences between the controllers in terms of plant OC. In the study, constant manipulated variables, simple PI and FF controllers, as well as a more advanced MPC controller, were evaluated. The controllers differed in the information used about the process and in the complexity of the control algorithms. For each control strategy, an operational map was constructed in order to obtain optimal set-point values.

The study shows that control strategies that use on-line nitrate and ammonia measurements give better performance with respect to OC than control algorithms with constant manipulated variables or oxygen control. They are also less sensitive to the optimal choice of controller set-points and can be valid for a wider range of plant operating conditions. While control strategies with more complex control structures and algorithms yield constant and significant improvement in plant performance with respect to effluent fines, they also result in higher OC (aeration costs, carbon costs) and therefore their implementation is not always justifiable. In particular, the study shows that with relatively simple PI nitrate and FF-PI ammonia controllers nearly similar OC can be achieved as with more advanced MPC algorithms. More advanced control algorithms are advantageous only when the plant is highly loaded and if stringent effluent fines are imposed by legislation. It is expected that in reality the differences between advanced MPC algorithms and relatively simple PI and FF control algorithms would be even smaller considering that an MPC with a perfect model, perfect measurements and perfect knowledge of the disturbances was used in the study. Hence, the improvement is thus more related to control structure, i.e. where the sensors and the actuators are located, than to the control algorithms.

Acknowledgement

The financial support of the Ministry of Higher Education, Science and Technology (P2-0001) is gratefully acknowledged.

REFERENCES

Ayesa, E., De la Sota, A., Grau, P., Sagarna, J.M., Salterain, A., Suescun, J., 2006. Supervisory control strategies for the new WWTP of Galindo-Bilbao: the long run from the conceptual design to the full-scale experimental validation. Water Sci. Technol. 53 (4–5), 193–201.

Carstensen, J., 1994. Identification of wastewater processes. Ph.D. Thesis, Institute of Mathematical Modelling, Technical University of Denmark.

Copp, J.B. (Ed.), 2002. The COST Simulation Benchmark—Description and Simulator Manual. Office for Official Publications of the European Communities, Luxembourg.

Henze, M., Gujer, W., Mino, T., van Loosdrecht, M., 2000. Activated Sludge Models ASM1, ASM2, ASM2d and ASM3. IWA Publishing, London, England.

Ingildsen, P., 2002. Realising full-scale control in wastewater treatment systems using in situ nutrient sensors. Ph.D. Thesis, Department of Industrial Electrical Engineering and Automation, Lund University, Sweden.

Jeppsson, U., 2005. Aeration and mixing energy for BSM1, BSM1_LT and BSM2. Benchmark Internal Report September 2005, available for download from http://benchmarkwwtp.org>.

Lindberg, C.F., 1997. Control and estimation strategies applied to the activated sludge process. Ph.D. Thesis, Department of Systems and Control, Uppsala University, Sweden.

Lukasse, L.J.S., Keesman, K.J., 1999. Optimised operation and design of alternating activated sludge processes for N-removal. Water Res. 33 (11), 2651–2659.

Maciejowski, J.M., 2002. Predictive Control with Constraints. Prentice-Hall, Harlow.

Olsson, G., Newell, B., 1999. Wastewater Treatment Systems, Modelling, Diagnosis and Control. IWA Publishing, London.

Seferlis, P., Georgiadis, M.C. (Eds.), 2004. The Integration of Process Design and Control. Elsevier, Amsterdam.

Steffens, M.A., Lant, P.A., 1999. Multivariable control of nutrient-removing activated sludge systems. Water Res. 33 (12), 2864–2878.

Takács, I., Patry, G.G., Nolasco, D., 1991. A dynamic model of the clarification thickening process. Water Res. 25 (10), 1263–1271.

Vanrolleghem, P.A., Jeppsson, U., Carstensen, J., Carlsson, B., Olsson, G., 1996. Integration of wastewater treatment plant design and operation—a systematic approach using cost functions. Water Sci. Technol. 34 (3–4), 159–171.

Vrečko, D., Hvala, N., Carlsson, B., 2003. Feedforward-feedback control of an activated sludge process: a simulation study. Water Sci. Technol. 47 (12), 19–26.

Yuan, Z., Oehmen, A., Ingildsen, P., 2002. Control of nitrate recirculation flow in predenitrification systems. Water Sci. Technol. 45 (4–5), 29–36.